CONTROLLING CHAOS:
GOVERNANCE AND REGULATION OF AI-CONTROLLED AUTONOMOUS WEAPON SYSTEMS

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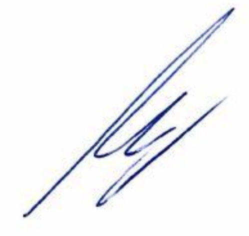
SIGNED:  **RANK:** LtCol

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ABSTRACT

Based on intensive literature research and the evaluation of case studies, this thesis first explains necessary definitions, such as human-in/on/out of the loop, in order to then assess the real danger posed by the deployment of artificial intelligence (AI)-controlled autonomous weapon systems (AWS). To this end, the thesis focuses on the tactical, operational, strategic, but also ethical implications for future warfare. In current conflicts, such as those in Ukraine and Gaza, the development of this technology is advancing at an ever-faster pace. Without meaningful human control, tactical advantages such as speed, precision and adaptive behaviour lead to significant weaknesses in ethical frameworks such as international humanitarian law (IHL). One of the biggest challenges is the accountability gap as there are no laws governing who is responsible for the misconduct of an AI-controlled AWS. At the strategic level in particular such misconduct can lead to an AI security dilemma, which could further fuel geopolitical conflicts.

The thesis aims to use this information and, by combining it with a critical analysis of existing laws and regulations and an assessment of current power dynamics, to develop options for governing and regulating AI-controlled AWS. Due to global tensions and disagreements over definitions, the international community has not yet succeeded in developing a binding set of laws based on ethical standards or at least adapting existing laws to the deployment of AI-controlled AWS. For this reason, this thesis presents a tiered governance model that links the autonomy levels of AI-controlled AWS with traditional levels of warfare. Together with recommendations for implementation, this pragmatic model is intended to serve as a temporary solution for ethically conscious armed forces until it can be replaced by a more comprehensive, ethically grounded legal framework, ideally within the framework of international humanitarian law.

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AI	-	Artificial Intelligence
AWS	-	Autonomous Weapon Systems
CCW	-	Convention on Certain Conventional Weapons
CSKR	-	Campaign to Stop Killer Robots
EU	-	European Union
FPV	-	First Person View
ICRC	-	International Committee of the Red Cross
IHL	-	International Humanitarian Law
LAWS	-	Lethal Autonomous Weapon Systems
NGO	-	Non-Governmental Organizations
NLP	-	Natural Language Processing
NPT	-	Treaty on the Non-Proliferation of Nuclear Weapons
OODA	-	Observe – Orient – Decide – Act
ROE	-	Rules of engagement
UAV	-	Unmanned Aerial Vehicle
UN	-	United Nations
UN GGE	-	United Nations Group of Governmental Experts
UNODA	-	United Nations Office for Disarmament Affairs
USV	-	Unmanned Surface Vehicle
WMD	-	Weapon of mass destruction
XAI	-	Explainable Artificial Intelligence

1 Introduction

“Machines can do many things, but they cannot create meaning. They cannot answer these questions for us. Machines cannot tell us what we value, what choices we should make. The world we are creating is one that will have intelligent machines in it, but it is not for them. It is a world for us (Scharre, 2018).”

This simple and noble idea seems subjectively correct, but reality has caught up with it in recent years. The integration of artificial intelligence (AI) into autonomous weapons systems (AWS) has been significantly advanced. Autonomous weapon systems, such as drones, have been created that are able to analyse situations in a fraction of a second and make an appropriate decision. Robotics and machine learning are thus changing the tactical and operational capabilities on the modern battlefield. However, without human intervention, this also creates complex legal and ethical dilemmas that fundamentally challenge conventional norms of warfare and deterrence. This has ushered in the beginning of a new era in warfare, which has serious implications for national and international security, global stability, but also for basic ethical understanding. It is therefore absolutely necessary to understand the use of this technology and the resulting ethical consequences in order to prepare, protect oneself, avoid mistakes and, if necessary, regulate the use of this technology.

The use of this technology is not only theoretically possible. Since Russia's large-scale invasion of Ukraine on 24 February 2022, there have been frequent and detailed reports on how this advanced technology is changing the way war is waged. For example, it is only the use of Unmanned Aerial Vehicles (UAVs) that has enabled Ukraine to defend itself for so long against Russia, which has superior numbers and technology. Amos Chapple states that drones are already being used not only for reconnaissance, supply or precision strikes, but also to disrupt enemy logistics and command structures far behind enemy lines. To do this, semi-autonomous drone swarms are used that are able to coordinate themselves within a battle network to fulfil predefined missions (Chapple, 2024). Even though these drone swarms are still controlled by one or more operators, most of the actions are already fully automated. This development has a significant influence on the battlefield in Ukraine and is an outlook on future conflicts.

The biggest advantage of this technology is also one of the biggest points of criticism. James Johnson argues that AI-supported technologies significantly accelerate the observe-orient-decide-act (OODA) loop and thus shorten tactical reaction time on the battlefield, but that they also increase the risk of misinterpretation. He argues that the technology, which is designed to optimize data processing and relieve the burden on humans, makes fewer mistakes than a human being, but due to the much larger amount of data processed, the risk is the same. Especially in conflicts involving nuclear-weapon states, such misinterpretation can lead to an unintended escalation. This raises the question of whether this technology can be trusted enough to be allowed to act independently. And if it comes to such an operation, with humans-out-of-the-loop, does this technology reflect the existential risks that arose from the nuclear arms race of the twentieth century (Johnson, 2019, 2020a)? Such an arms race could further deteriorate the already tense global stability by causing states to feel compelled to use untested AI-controlled systems for fear of technological inferiority. As early as 1950, John Herz coined the term "security dilemma" to describe the danger of a nuclear arms race (Herz, 1950). James Johnson and Vincent Boulanin take up this idea and expand the term to the "AI security dilemma." Just as Herz assumed as early as the 1950s that nuclear armament for defensive purposes could be misinterpreted as an aggressive act, Johnson and Boulanin transfer this danger into the 21st century (Boulanin, 2020; Johnson, 2020b). Are the mistakes of history are repeated by using AI-controlled autonomous defensive systems to develop autonomous AI-controlled AWS? And is this threat really comparable to the nuclear threat of the Cold War?

What is certain is that the technology of AI-controlled AWS is fundamentally changing modern warfare. It not only requires a change in established military doctrines but also challenges the concept of deterrence and escalation control. Traditionally, the concept of deterrence is based on the predictable actions of human actors. In the future, this concept will have to be extended to AI systems that are also capable of making independent decisions. Andrew Futter has found that it is precisely these independent decisions that make it very difficult for humans to assess them, as this is usually based on knowledge, experience and intuition. This creates the need to use AI to evaluate an action, which in turn increases the risk of misinterpretation. This is because when a system is driven by algorithmic analysis rather than human intuition, it can react unpredictably in high-pressure situations. This unpredictability undermines the stability on which deterrence theory relies (Futter, 2022).

The deployment of AI- controlled systems and AWS not only requires a rethink of tactical and operational warfare, as well as national and international security strategies, but also challenges ethical principles. One term that is increasingly being used by ethic scholars is the so-called “gamification” of warfare. This term describes the reduction of a complex warfare to data and simulations. As Dominika Kunertova warns, relying on AI for warfare could trivialize the human cost of conflict by framing military operations as algorithmic exercises rather than ethically significant decisions. This could lower the threshold for the use of force as policymakers may become desensitized to the consequences of military action (Kunertova, 2023). But soldiers are also affected by this trivialization, in that the decision about life and death is not made by them, but by an AI. Stuart Russell argues that delegating life-and-death decisions to machines undermines human responsibility and ethical accountability. Without human conscience and scruples, which are absent in human-out-of-the-loop AWS, unintended civilian casualties are accepted, undermining basic principles of international humanitarian law. In particular, the principles of distinction, proportionality and humanity in the Geneva Convention must be questioned, because they assume that people, not machines, make critical decisions. AI- controlled systems that lack an understanding of context, cultural nuances, or moral complexity may have difficulty distinguishing between combatants and non-combatants. This limitation raises the question of whether this technology can be considered dangerous enough to adapt existing regulatory frameworks, such as the Treaty on the Non-Proliferation of Nuclear Weapons (NPT), to meet the unique challenges of AI- controlled AWS (Russel, 2023).

With intensive literature research, the evaluation of case studies, a critical analysis of existing laws and regulations and an assessment of current power dynamics, this thesis aims to answer three main questions:

1) How advanced and dangerous are AI-controlled AWS really?
2) How will the deployment of AI-controlled AWS change future warfare?
3) How can the deployment of AI-controlled AWS be regulated?

By answering these questions, regulatory measures can then be developed in the final section of this thesis to close the gaps in existing international laws and to address the risks associated with the deployment of AI-controlled AWS.

The deployment of AI-controlled AWS is not just a technological milestone, but a paradigm shift that, just like nuclear technology, requires a careful consideration of its legal, strategic, but also ethical implications. The results of this thesis can be incorporated into ongoing and upcoming debates on this topic to underscore the urgency of a solid regulatory framework.

2 AI- Controlled Autonomous Weapon Systems

The deployment of AI-controlled AWS is fundamentally changing modern warfare. In recent years, and driven by current conflicts, simple drones as unmanned aerial vehicles (UAVs) have evolved from reconnaissance tools to fully integrated combat systems that are even capable of autonomously detecting and attacking targets. Advanced AI-controlled AWS are able to operate at unprecedented speeds, processing battlefield data in real time and making targeting decisions in fractions of a second. While these capabilities offer significant advantages, such as reducing the loss of life among soldiers, they also lead to new vulnerabilities and risks (Chapple, 2024). For example, in addition to the ever-growing cyber threat, one of the most pressing challenges is the decreasing human oversight of AI decision-making in critical situations (Boulanin, 2020; Johnson, 2020a). While semi-autonomous AWS still rely on an operator to authorise critical actions, fully autonomous AWS can select, and attack targets based on pre-programmed algorithms and machine learning models (Scharre, 2018). This transition from human-on-the-loop to human-out-of-the-loop systems therefore raises important questions about legal and ethical dilemmas, including compliance with international humanitarian law, accountability in warfare, and the potential for unintended escalations (Docherty, 2012; ICRC, 2021).

This Section uses case studies to examine the current state of AI- controlled AWS technology, defines key concepts, and assesses both the military advantages and the associated risks. In addition, the danger posed by AI-controlled AWS is examined and evaluated in more detail. Understanding these elements is crucial to assessing the strategic, legal, and ethical challenges that AI-controlled AWS pose to modern warfare.

2.1 Definition

AI-controlled automatic weapon systems (AWS), which include drones, are fundamentally changing modern warfare. While conventional weapon systems require human control or at least human intervention (Horowitz, 2019), AWS can use machine learning and advanced algorithms to identify and track targets with minimal or no human supervision. Lethal autonomous weapon systems (LAWS) are even capable of attacking targets fully autonomously (Johnson, 2019). The necessary decisions are made by the AWS in just a fraction of a second. The increasing deployment of and dependence on AWS has triggered an intensive debate about the definition, legitimacy and ethical implications for military operations and broader security frameworks (Boulanin, 2020). While the term LAWS is commonly used in international legal debates, this thesis adopts the broader term AWS. This allows for the inclusion of both lethal and non-lethal systems with varying levels of autonomy, reflecting the full spectrum of technological and regulatory challenges posed by AI-controlled AWS in modern warfare.

A definition of AI-controlled AWS is controversial in both academic and military circles. For instance, the US Department of Defense defines AWS as “weapons that, once activated, can select and engage targets without further intervention by a human operator” (US DoD, 2023). Paul Scharre and Michael Horowitz use a similar definition in their article: “A weapon system that, once activated, is intended to select and engage targets where a human has not decided those specific targets are to be engaged” (Scharre and Horowitz, 2015). In both cases, it can be criticized that these definitions do not sufficiently address the distinction between semi-autonomous and fully autonomous systems, nor do they take into account defensive or offensive use. However, these factors are of crucial importance for ethical and legal classification. As the largest international organisation, there is no generally recognised definition of AWS even in the United Nations (UN). However, the United Nations Office for Disarmament Affairs (UNODA) recognises the importance of a common definition and is promoting further discussions between member states (UNODA, 2023). The lack of a generally recognised definition of AWS allows states to exploit regulatory gaps and to develop these systems without standardised guidelines for transparency and accountability.

Human Rights Watch addressed this issue in a comprehensive article as early as 2012 and introduced a structured classification of the levels of automation, which is still up to date and continues to be used in many articles:

1) **Human-*in*-the-Loop Weapons**: Robots that can select targets and deliver force only with a human command.

2) **Human-*on*-the-Loop Weapons**: Robots that can select targets and deliver force under the oversight of a human operator who can override the robots’ actions.

3) **Human-*out-of*-the-Loop Weapons**: Robots that are capable of selecting targets and delivering force without any human input or interaction.

(Docherty, 2012)

Understanding these levels of autonomy is essential to assess the risks associated with these systems and to develop regulatory mechanisms that address both ethical concerns and operational challenges.

With this definition, it is possible to look at these three levels from a technological perspective and to identify the critical functions of an AWS:

1) **Sensor fusion**: Robots with integrated radar, camera and infrared sensors to detect and track targets. (Shaw, 2024)
2) **Machine learning algorithms**: Robot models that analyse battlefield data to predict enemy movements and optimise target decisions. (Verbruggen and Boulanin, 2017)
3) **Autonomous decision-making**: Robots with the ability to process large amounts of information in real time and combat threats autonomously. (Russell, 2019)

These two types of definition are not so different, but they do show that there is a fundamental tension between ethical accountability (definition according to Human Rights Watch) and operational efficiency (definition according to critical functions). Until these tensions are resolved, the international community remains divided on whether AWS should be regulated in a manner similar to nuclear weapons, or whether their development can be managed through legal and ethical restrictions rather than outright bans (Noor, 2023; UNODA, 2023).

After analysing existing definitions, it turns out that the highly complex technology of AI-controlled AWS cannot be given in just a few words. A definition system is needed that covers all aspects of this technology and deals with at least the following points:

1) **Autonomy Levels:** Differentiating between AWS that assist human decision-making and those that operate entirely independently.

2) **Lethality and Engagement Criteria:** Addressing whether a system is designed for defensive or offensive purposes and whether it meets international law standards.

3) **Level of AI Integration and Adaptability:** Recognizing the extent to which machine learning and autonomous decision-making are embedded into the system's operational framework.

4) **Accountability Mechanisms:** Ensuring that AWS deployment includes oversight structures to prevent legal and ethical violations.

These points could be used to create a system of definitions that closes existing conceptual gaps by encompassing the diverse functions of AWS and at the same time taking into account the ethical and regulatory challenges. However, it is imperative that the international community solves these challenges and clarifies the resulting questions by mutual agreement, because AI-controlled AWS will not only change the way wars are fought in the future, they are changing them today.

2.2 Case Studies

The latest military conflicts have driven the development of advanced military technology. This includes the integration of AI into AWS, which are already being used in various conflicts, but with different levels of autonomy and capability.

One such conflict is the ongoing war between Ukraine and Russia, in which both nations are making extensive use of AI-supported and -controlled drones. Ukraine, for example, is making very successful use of small first-person-view (FPV) drones for surveillance and reconnaissance, but also for targeted attacks. These drones work with AI-based navigation and targeting systems that allow for minimal human intervention. Even when deployed in situations where electronic warfare causes signal interference, these drones can still navigate, steer themselves to the target and thus complete the mission (Jacoby, 2024).

In addition to the increased use of autonomous unmanned aerial vehicles (UAVs), Ukraine is also pioneering the use of robot-assisted ground units. In December 2024, Ukraine successfully deployed a platoon-sized unit of robots equipped with heavy machine guns with AI-based targeting systems in an operation in the Kharkiv region. These robots have demonstrated the ability to defend themselves and lay anti-tank mines despite being damaged and are expected to reduce the risk to soldiers on the front line in the future. Even though these robots were controlled by operators from the 13th “Khartiia” brigade, this deployment highlights a significant shift towards automated warfare and underscores the potential of AI-supported ground units in modern military operations (Serohina, 2024; Hnidyi, 2025).

In addition, Ukraine has successfully deployed AI-controlled unmanned surface vessels (USVs) against Russian installations and ships in the Black Sea. These drone boats were equipped with explosives and are able to navigate independently to their target without an operator. The use of relatively inexpensive USV poses a significant risk to the strong Russian Navy and reflects a broader shift towards the use of AI-controlled AWS for maritime dominance (Keller, 2025).

For the examples shown above, the use of drones in all three domains still required an operator, at least a human-on-the-loop AWS. Both Ukraine and Russia are currently developing and testing AI-controlled drones with humans-out-of-the-loop. These drones are able to identify and attack targets independently (Chapple, 2024). However, this type of AWS has not only been used since the Ukraine war. According to a report by the United Nations, the use of an AI-controlled AWS was documented in 2020 during the civil war in Libya. In the process, Turkish-made Kargu-2 drones targeted and attacked retreating troops without direct human control. This event is considered one of the first documented instances of human-out-of-the-loop AWS and triggered fierce debates, particularly about compliance with international humanitarian law in the decision-making process of an AI when it comes to life and death (ICRC, 2024).

The AI-controlled AWS first used in Libya and currently in the Ukraine war is referred to as loitering munition. Another frequently used variant is the IAI Harpy developed by Israel Aerospace Industries. With this AWS, humans are no longer involved, and they independently recognise and identify targets that emit radar signals and automatically attack them (IAI, 2021). Although they are often presented as defence systems, their ability to operate independently of human supervision highlights the risks of automation bias and unintended escalation particularly in complex combat environments where misidentification of targets could lead to strategic miscalculations (Johnson, 2020b).

The examples listed here show that the level of autonomy in AI-controlled AWS varies depending on the application. For the most part, AI has been integrated into many military systems to automate certain functions and thus exploit operational advantages such as faster decision-making and lower risk for human soldiers. The challenges that have not yet been resolved, such as ensuring compliance with international laws and taking ethical concerns into account when making life-and-death decisions, still make it necessary to involve human control in critical actions. Nevertheless, AWS without human operators have already been deployed and the development of this technology is advancing rapidly. In this context, not only technological limitations and threats must be overcome, but military efficiency must also be reconciled with ethical responsibility.

2.3 Technical Limitations and Risks

Even if AI-controlled AWS are already in use and development is progressing rapidly due to current global conflicts, there are still significant technical limitations. These limitations minimise reliability and effectiveness in a complex combat environment and lead to significant military and ethical risks.

A basic requirement for the proper functioning and a high level of operational effectiveness of AWS controlled by AI is continuous and correct sensor input, e.g. from infrared cameras, radar and microphones. Computer vision algorithms are dependent on the sensors and can therefore be unreliable in bad weather conditions, on uneven terrain or when electronic countermeasures are used. Even if sensors are still more reliable than human judgement under poor environmental conditions, these disruptive factors can still lead to the same misjudgements. For example, research has shown that AI-controlled AWS have difficulty distinguishing between combatants and civilians when confronted with deceptive or confusing environments (Verbruggen and Boulanin, 2017). A relevant example is the use of AI-controlled drones in the war in Ukraine. After Russia actively restricted the Ukrainian use of AI-controlled drones by disrupting the GPS network, Ukraine adapted its navigation system. By integrating an AI-controlled navigation system, the use of drones is, at least theoretically, possible even in signal-free spaces and without a connection to the operator (Jacoby, 2024). Russia, in turn, has countered this development by introducing false data to make the drones target the wrong places or targets, for example. This heavy reliance on sensor data clearly demonstrates one of the vulnerabilities of AI-controlled decision-making (Johnson, 2020a).

AI decision making is not based on human intuition and experience, but solely on logical probabilistic assessments. As shown, these probabilistic assessments can also be flawed. Another major risk, however, is the so-called automation bias. This is when human operators place too much trust in AI-made decisions without critical supervision (Bode and Bhila, 2024; Hoffman, 2024). Studies have shown that operators under high pressure in combat situations rely too heavily on AI-generated target recommendations, possibly leading to unlawful action (Johnson, 2022). The incident in Libya in 2020, in which a Kargu-2 drone attacked a target unmanned and independently, serves as a practical example (ICRC, 2024). This vulnerability raises the question of whether AI-controlled AWS can and should be trusted, since, just like humans, they make mistakes, but their decisions do not include morals and ethics.

Just as AI-controlled AWS are dependent on sensor data, they are also highly dependent on networked communication and real-time data processing. This makes them a vulnerable target for cyber threats. Experts warn that AI-controlled AWS can be hacked, disrupted or their commands falsified to cause unintended attacks or system failures (Verbruggen and Boulanin, 2017). In the context of the war in Ukraine, both Ukraine and Russia are using electronic warfare techniques to disrupt and disable AWS. There have been reported incidents in which Ukrainian drone operators have lost control of AI-based drones after Russian electronic countermeasures have disrupted or hijacked their communication link. These drones then crashed or returned to base (Freedberg, 2023; Panella, 2025). This illustrates the precarious nature of AI-supported or -controlled AWS operating in contested digital spaces where hostile interference can make these systems unpredictable and even dangerous.

The further the development of AI-controlled AWS technology advances, the more accessible it becomes and, accordingly, the risk of unwanted proliferation also increases. Non-state actors, such as terrorist organisations or paramilitary groups, could acquire AI-controlled AWS and use it for their own purposes (UNGA, 2024). An example is the potential misuse commercially available drones for offensive operations by the Islamic State (IS). The IS misused commercial drones to carry explosives and attacked military targets in Iraq and Syria with them. Especially during the battle for Mosul in 2016 and 2017, these armed drones were used to drop grenades and other munitions, what caused significant casualties and damage (Brown, 2020). Another report describes the attack by Houthi militants on merchant ships in the Red Sea. They are increasingly using a combination of rocket and drone attacks. In the most recent attacks, they have not only used UAVs but also USVs (Saul and Maltezou, 2024). Even if drones are used on a large scale in asymmetric warfare, there is currently no evidence that they are supported or even controlled by AI. Nevertheless, the risk will increase over time until non-state actors integrate AI into their weapons systems, since AI is already being used in other areas, such as recruitment or propaganda (Nisar, 2023; Nelu, 2024).

The use of drones by non-state organisations or the military use of AI-controlled commercial drones in the Ukraine war also highlights another challenge in dealing with AWS: determining accountability. In the event of an unlawful attack by an AI-controlled AWS, is accountability borne by the programmer, the manufacturer, the commanding officer, the state or all of them together (Verbruggen and Boulanin, 2017)?

While traditional military structures are based on the responsibility of a command, this principle is complicated when using AI-controlled AWS, since the decision-making processes involve only partial or even no human oversight. There is a “accountability gap” in the use of AWS. AI-controlled AWS lack an ethical and moral capacity to act, so they are not subject to the same legal standards as human soldiers (Docherty, 2015; Horowitz, 2016; ICRC, 2021). The problem is exacerbated by the automation bias described above. Even if a human operator supervises the actions of an AI-controlled AWS, the automation bias can cause the operator to subordinate AI decision-making (Horowitz, 2016). As long as there is no clear legal framework for the use of AI-controlled AWS, there is always a risk of creating situations in which accountability is diluted. This makes it difficult to hold actors accountable for violations of the principles of “jus in bello”, such as distinction and proportionality (Johnson, 2022).

2.4 Evaluating the Real Threat of AI-Controlled AWS

The integration of AI into military conflicts and operations is often compared to other historical technological revolutions, such as the development of nuclear weapons or the beginning of cyber warfare. According to scientists like James Johnson or Paul Scharre, AI-controlled AWS represent a fundamental change in military strategy similar to that of the emergence of nuclear deterrence (Scharre, 2018; Johnson, 2024).

One of the strongest arguments against the use of AI-controlled AWS is that they have the potential to accelerate conflicts beyond human control. The fact that AI-based and controlled systems, including AWS, can process huge amounts of data much faster than humans can, significantly accelerates the observe-orient-decide-act (OODA) loop. On the one hand, this increases tactical and operational efficiency, but on the other hand, it also leads to cascading strategic instability, as the time available for decisive de-escalation is constantly being reduced. An unintended escalation that arises in this way could ultimately lead to nuclear war (Johnson, 2020a, 2022). In addition to the significantly accelerated OODA loop, James Johnson warns of the danger that AI-controlled systems could make wrong decisions in crisis scenarios due to misinterpretation of data, confusing harmless activities with hostile intentions. As an example, he cites the historic incident of the Soviet nuclear false alarm of 1983, when the Soviet early-warning system mistakenly detected an incoming U.S. missile attack. It was only thanks to the intuition and scepticism of the Soviet officer Stanislav Petrov that the Soviet counterstrike was avoided. If an AI-controlled system had been involved, it might have come to a nuclear counterstrike on the basis of logical probabilistic assessments (Johnson, 2024).

In connection with autonomous deterrence systems, Paul Scharre discusses the risks associated with delegating the life-death decision to machines. He explains that AI has no understanding of context, diplomacy or human intentions, which makes it a poor substitute for human commanders in high-risk military decision-making. Since nuclear weapons still have to be deliberately activated by humans, the risk of accidental escalation currently only plays a role with AI-controlled AWS, since these can automatically respond to perceived threats (Scharre, 2018). However, this could change quickly in the future due to a “security dilemma”. As early as 1950, John Herz coined the term “security dilemma” to describe the danger of a nuclear arms race (Herz, 1950). James Johnson and

Vincent Boulanin take up this idea and expand the term to the “AI security dilemma.” Just as Herz assumed as early as the 1950s that nuclear armament for defensive purposes could be misinterpreted as an aggressive act, Johnson and Boulanin transfer this danger into the 21st century. If a state fears that an adversary's AI capabilities are superior, it could feel pressured to develop autonomous first-strike options, which would further erode strategic stability (Boulanin, 2020; Johnson, 2020b).

If AI-controlled AWS really pose an existential threat in the future, similar to the nuclear threat, what defence measures can be used to mitigate this risk? The best and most immediate protection is and remains the continued presence of human supervision to veto AI-made decisions in critical situations. This means that an exclusive restriction to human-in-the-loop and human-on-the-loop systems is necessary (Docherty, 2012), which would mean that the tactical and operational advantage of human-out-of-loop systems would be lost. And even then, there is a risk that this human supervision could become increasingly superficial rather than meaningful due to automation bias (Bode and Bhila, 2024).

Another defence measure is the vulnerability of AI-controlled AWS to cyber attacks and electronic warfare. With reports from the Ukraine war, it has already been proven that these methods can be used to disrupt, crash or reverse enemy AI-controlled AWS (Freedberg, 2023; Panella, 2025). Just as the methods of manipulation continue to evolve, so do the defence measures of the AI-controlled AWS, which minimises the chances of success. In addition, there is a risk that the hijacked AI- controlled AWS attack false and unintended targets due to corrupted data (Scharre, 2018).

One sure way to defend against this would be to regulate the use of AI-controlled AWS before they are deployed. The UNODA has proposed various frameworks for regulation, but unlike the Treaty on the Non-Proliferation of Nuclear Weapons (NPT) for nuclear weapons, there is currently no legally binding global treaty for AI-controlled AWS (UNODA, 2023). Some authors argue that nuclear weapons are inherently easier to regulate than AI-controlled AWS because of their massive destructive power. The reason is that they are easily accessible and can be developed incrementally by many actors, even non-state groups (Verbruggen and Boulanin, 2017; Scharre, 2018; Johnson, 2020b, 2020a). It is therefore reasonable to assume that the proliferation of AI-controlled AWS is more difficult to contain than that of nuclear weapons. Despite these concerns, AI-controlled AWS are currently not capable of strategic nuclear decision-making. They primarily influence tactical and operational warfare rather than strategic deterrence. However, due to the ever-accelerating pace of technological development and a lack of a binding legal framework, there is a risk that this will change in the future.

2.5 Conclusion

This Section has attempted to analyse and describe the danger posed by AI-controlled AWS. It has been found that the factor of effectiveness is not sufficient on its own, but that autonomy, unpredictability, accountability and the potential for proliferation must also be taken into account.

While traditional weapon systems serve as a pure tool for a responsible human utilisation decision, an AI-controlled AWS fundamentally changes this concept. By transferring life-and-death decisions to an AI, not only are existing military capabilities greatly improved, but the nature of modern warfare is qualitatively changed. As a consequence, the risk of unpredictability of the behaviour of AI-controlled AWS in complex operational environments arises. Even the most advanced AI-controlled AWS are still too dependent on sensor data and lack context awareness, moral judgement and the ability to distinguish with certainty between legitimate and unlawful targets (Johnson, 2022). Current AI-controlled AWS base their decisions on pattern recognition, data correlation and probabilistic analysis, which, just like humans, makes errors inevitable in critical situations. However, unlike humans, AI-controlled AWS make these mistakes with machine speed and scale, making human interaction and interruption almost impossible. In this way, misidentification of threats or targeting errors can occur, which in turn can lead to civilian casualties, friendly fire or even an escalation of a conflict (Bode and Bhila, 2024).

In addition to the use of AI-controlled AWS by states in current conflicts, the proliferation of this technology poses a new threat. Unlike nuclear weapons, which require state-controlled infrastructure, rare materials and extensive expertise, AI-controlled AWS can be developed and adapted step by step using commercial AI and robotic technologies.

This makes their procurement and use easier and more attractive for non-state actors, such as terrorist organisations and insurgent groups (Scharre, 2018). Even if there is currently no confirmed use of AI-controlled AWS by non-state actors, the asymmetric use of this technology, for example for autonomous assassinations, is an imminent reality.

Another critical issue is the lack of accountability. The current legal framework is unclear and cannot be applied when AI-controlled AWS are used. This ambiguity creates a dangerous legal vacuum in which AWS can be used without the same legal control as human soldiers (ICRC, 2021). This gap in accountability could tempt states to use AWS in legally and ethically questionable scenarios, further complicating conflict resolution.

Finally, the positive arguments that AI-controlled AWS will reduce the number of casualties at the front and increase precision are not wrong in principle, but they must be considered in relation to their reliability, control and effectiveness on a highly intense and competitive battlefield. The ethical and legal consequences must not be neglected. Yet as history has shown, military technology often develops faster than the ethical and legal systems designed to control it. AI-controlled AWS, once deployed at scale, could give rise to new forms of warfare that humanity is unprepared for. It depends on whether states act now to regulate them or allow their development to spiral out of control.

3 The Future Impact of AI-Controlled AWS on Warfare

The rapid development and deployment of AI-controlled AWS are already influencing current conflicts, but their impact on future warfare will be even more significant, as this change will go beyond mere automation. The primary aim of using AI-controlled AWS is to accelerate the OODA cycle, improve situational awareness and increase operational capability in contested environments (Scharre, 2018; Johnson, 2020a). In order to effectively leverage these advantages, the resulting challenges, such as unpredictable AI behaviour and vulnerability to cyber attacks, must be overcome (Verbruggen and Boulanin, 2017). In addition, traditional command and control structures must adapt as AI increasingly replaces or supplements human tasks in decision-making (Podar and Colijn, 2025).

This Section seeks to answer the question of how exactly the use of AI-controlled AWS will change future warfare. To this end, the analysis focuses on the traditional levels of warfare and examines how AI-controlled AWS will influence decision-making, power projection, deterrence and military doctrine. The aim is not only to understand the advantages of these technologies, but also to critically assess their risks and the potential consequences of their widespread implementation.

3.1 Future Tactical Implications of AI-Controlled AWS

In the future, AI-controlled AWS will exceed the military capabilities of humans, such as target -identification or -prediction, by not only having access to a battlefield network but also being able to process its data in real time. In doing so, advanced machine learning models will not only analyse sensor fusion data, but also compare it with integrated infrared, radar and electromagnetic signals to further improve situational awareness and precision on the battlefield (Nair, 2024). As discussed in Section 2.4, one of the biggest deployment risks remains the vulnerability to hostile manipulations, such as interference and data poisoning attacks, designed to mislead the AI-controlled AWS. To mitigate this risk, the further development and integration of multimodal AI models that link multiple input sources before targeting a goal is necessary. This reduces the effect of interference and decreases the likelihood of misidentification (Verbruggen and Boulanin, 2017).

Even if this integration may slow down the reaction speed of AI-controlled AWS, they are and will remain capable of making decisions at machine speed and significantly accelerating the OODA loop compared to humans. This makes them particularly effective in future warfare in defence against modern weapon systems such as hypersonic threats or drone swarms (Horowitz, 2019). But even if electronic warfare threats can be successfully averted, it is precisely this speed of decision-making that leads to the risk of unintended escalation, as ambiguous threats could be misinterpreted without human control or supervision (Johnson, 2020a). To prevent this, explainable AI (XAI) could be implemented, which can provide real-time justifications for targeting decisions, enabling human operators to review and override AI-controlled AWS actions when necessary (Wood, 2024). Ultimately, this requires the implementation of kill switches and override mechanisms that enable operators to stop AWS operations in real time if errors or anomalies are detected (Etzioni and Etzioni, 2017). In addition, the development of constraint-based AI models could be expedient, which use predefined ethical parameters to restrict the actions of AI-controlled AWS and prevent unwanted operations (Hellström, 2013). However, the most effective security mechanism in the future will continue to be the use of human operators. But even if human verification checkpoints in human-in-the-loop systems can reduce unintended engagements, they may also introduce delays or decrease algorithmic efficiency in high-speed operations, depending on the tactical context and the level of automation (Sele and Chugunova, 2024). The future use of AI-controlled AWS must therefore weigh up security and effectiveness, which is no different from the use of today's weapon systems.

AI-controlled AWS will be able to change their behaviour on the battlefield based on real-time tactical conditions by using adaptive reinforcement learning. This will allow them to adjust their probabilistic threat assessments to increase their effectiveness against a dynamic adversary. This adaptability leads to unconventional tactics and strategies that are unpredictable and contrary to standard military doctrine (Bode and Bhila, 2024). Especially in complex combat situations in which electronic warfare is used, the AI-controlled AWS is forced to constantly recalibrate its decision models (Freedberg, 2023). One of these situations is urban combat, which is inherently difficult and extremely challenging. Even soldiers have problems to locate the enemy and distinguishing between enemy and non-enemy forces in densely populated areas. While current AI-based detection systems have difficulties with contextual understanding despite a multitude of sensor data, natural language processing (NLP) will need to be integrated into future systems to interpret human behaviour and interactions and improve their ability to distinguish between threats and civilians (Hagos, Alami and Rawat, 2018). NLP will not only help to better identify potential targets, but also to improve communication with human operators, especially in mixed formations. In addition, geofencing can be used to further reduce the risk of confusion by defining fixed operational areas and no-go zones (Vagal, Markantonakis and Shepherd, 2021).

Future AI-controlled AWS will likely operate in coordinated formations with minimal human intervention and function as coherent combat units rather than as individual systems (Scharre, 2018). The development of the Skyborg program by the U.S. military illustrates this shift, in which autonomous combat aircraft work together with manned combat aircraft and act as force multipliers (Gunzinger and Autenreid, 2020). However, similar advancements in swarm intelligence allow AI-controlled AWS to share battlefield data and optimize collective engagement strategies (Verbruggen and Boulanin, 2017). Blockchain-based command authentication, which ensures that AWS follow only verified operational directives (Rashid and Khan, 2022), and decentralized AI control mechanisms, which reduce reliance on a single command hub, enabling AI- controlled AWS units to operate independently, even if their networks are compromised (Jacoby, 2024).

Current developments show that the integration of AI-controlled AWS into fully autonomous combat units is not just a theory, but a highly likely and two- edged future. Even if these systems promise greater precision, faster response to threats and adaptable strategies on the battlefield, they also generate new and very complex challenges. In the future, modern armed forces will have to weigh up whether the advantages outweigh the challenges in terms of reliability, ethical decision-making and escalation control. However, the question arises as to whether it is still a choice at all when human soldiers can hardly counter the unpredictable behaviour and rapid response of AI-controlled AWS in tactical warfare. Ultimately, the success of AWS in tactical warfare will depend on striking the right balance between autonomy and accountability to ensure that these systems serve as multipliers rather than uncontrollable escalators.

3.2 Future Operational Implications of AI-Controlled AWS

3.2.1 Command and Control

Traditional operational command and control (C2) structures are based on a hierarchical model in which command levels have different decision-making authorities. With the further development and future use of next-generation AI-controlled AWS, it is imperative that these operational C2 structures be adapted to take into account the decentralization of decision-making. This new structure must be able to integrate AI-controlled AWS that make decisions at machine speed in a way that does not create bottlenecks in the approval chain (Black *et al.*, 2024). In the future, it would be technically possible to use human-out-of-the-loop AWS without human supervision to circumvent this challenge. However, as discussed in Section 2.5, while these systems significantly accelerate the OODA loop, they are also prone to misinterpreting ambiguous data, which can lead to unintended escalations (Johnson, 2024). Furthermore, adversaries can exploit AI-dependent command networks through cyber operations, jamming or spoofing attacks to disrupt AWS decision-making or manipulate their behaviour (Podar and Colijn, 2025). For this reason, a resilient C2 structure is needed that incorporates XAI to provide human commanders with transparent justifications for AI-made decisions and to ensure that human oversight remains effective even in complex combat environments (Wood, 2024). This hybrid approach is emerging as a responsible solution for such a resilient C2 architecture because it combines decentralized AI decision-making with human oversight.

One approach to such a C2 structure currently being explored is cognitive teaming, in which AIs can operate as independent units on the battlefield but remain tethered to human control and instruction at key decision points (Ferl, 2024). The US military's Skyborg program, in which AI-controlled UAV work alongside human pilots, serves as a small-scale example of how AI can reduce cognitive load while maintaining operational coherence (Gunzinger and Autenreid, 2020).

3.2.2 Force Projection and Sustainment

The future increase in the use of AI-controlled AWS will significantly change the force projection and sustainment at the operational level and reduce the logistical effort required to deploy and maintain combat effectiveness. While armed forces currently rely on very extensive supply chains to provide personnel, materiel and ammunition to maintain combat effectiveness, for example, that AI- based or -controlled AWS can operate with minimal logistical support and extend the time that armed forces can be deployed in contested areas.

As described in Section 2.3, drone swarms and autonomous ISR systems are already being used in current conflicts, such as the war in Ukraine, to carry out reconnaissance and attack operations (Chapple, 2024). On the one hand, their precision and effectiveness reduce the consumption of ammunition, and on the other hand, fewer personnel are required for maintenance and operation than for modern alternative weapon systems, such as fighter jets or battle tanks. However, operational sustainability is limited by energy and computing constraints (Verbruggen and Boulanin, 2017). AI-controlled AWS require constant power and advanced computing capabilities. This makes not only the AI-controlled AWS vulnerable to electronic warfare and to cyber attacks, but also the essential supply chain (Rajabi, Beigi and Aghakhani, 2022). If AI-controlled AWS are compromised during transport or maintenance, adversaries could pre-program malware or latent kill switches, rendering them unreliable in combat (Simmons-Edler *et al.*, 2024). To counter this new challenge to sustainability in AWS logistics, cybersecurity could be extended to supply chains, for example. Additionally, blockchain-based autonomous authentication systems should be integrated to ensure integrity in battlefield AI-based logistics and prevent data tampering or unauthorized changes in AI-controlled AWS (Rashid and Khan, 2022).

3.2.3 Multi-Domain Integration

Even if the tactical use of AI-controlled AWSs in future conflicts is very effective on its own, the challenges outlined in Section 2.4 can only be overcome by fully integrating them into multi-domain operations. This integration will enable AI-controlled AWS to function as specialized autonomous combat units, in which they are connected to each other via networked combat systems, rather than just operating over land, air and sea. As an example, not only the use of UAVs, but also the increasing use of AI-controlled USVs in maritime operations. This mixed use of AI-controlled AWS from two domains is supported by a satellite-based information network and protected against cyber attacks by digital protective measures. This enables permanent surveillance of the maritime operational area, autonomous mine clearance and precision strikes over long distances (Keller, 2025). When conducting such multi-domain operations, the deployed AI-controlled AWS actively contribute to real-time intelligence ecosystems on the battlefield, enabling commanders to conduct precision-guided operations on a scale never before seen (Shaw, 2024).

Interoperability will continue to be a significant challenge. Not only the AI-controlled AWS from the various domains, but also those of coalition partners, must be able to exchange sensor data, threat assessments and target information in real time (Atkinson, 2024). Latency and compatibility issues between AI-controlled AWS and conventional military platforms can lead to operational friction and require the establishment of standardized machine-to-machine communication protocols. This is particularly important to enable human operators and commanders in multinational and multi-domain operations to intervene quickly in critical situations without causing a technical slowdown in OODA loop.

3.3 Future Strategic Impacts of AI-Controlled AWS

3.3.1 Evolution of Strategic Decision-Making

Another change that the use of AI-controlled AWS will bring is strategic decision-making. While traditional military strategies are developed based on human intuition, experience and diplomatic engagement, future strategies will be based on AI-based predictive threat analysis (Wood, 2024). In this context, more and more decisions for high-risk scenarios will be delegated to AI-controlled AWS in order to be able to counter the significantly reduced reaction time of opposing AI-controlled AWS (Johnson, 2024).

The resulting dependency on ever-faster reaction times and decision-making by AI-controlled AWS also increases the risk of miscalculations, especially in crisis scenarios involving nuclear-weapon states. AI-controlled AWS integrated with early warning systems and controlling missile defence systems could misinterpret signals of an adversary's intent and trigger pre-emptive military action before diplomatic de-escalation is even possible (Johnson, 2020a). While this risk cannot be eliminated, it must at least be mitigated by adjusting command structures and ensuring that human oversight remains effective even in complex combat environments (Ferl, 2024).

3.3.2 Deterrence and Stability

Section 2.5 showed that AI-controlled AWS challenge the traditional concept of deterrence and escalation control. Conventional deterrence theories are based on the assumption that rational human actors make predictable decisions based on cost-benefit analysis (Boulanin, 2020). Especially when nuclear weapon states are involved, credibility and predictability are important factors in fulfilling the principle of deterrence of mutually assured destruction (Herz, 1950). Although the Section 3.1 is limited to the tactical level, the unpredictability described there in the use of AI-controlled AWS can also be applied to the strategic level if the AI has control over autonomous launchers for nuclear weapons. In this case, the concept of deterrence is no longer based on human psychology, but on algorithmic logic that may not align with traditional strategic calculations (Johnson, 2020b).

In addition to this threat, the “AI security dilemma” may lead to a further escalation in the future if states that fear the superiority of opposing AI-controlled AWS may rush to deploy their own AI-controlled AWS, reminiscent of the nuclear arms race during the Cold War (Johnson, 2020b). This dynamic could destabilize regions where the proliferation of AI-controlled AWS occurs unchecked, as states could perceive defensive deployments as offensive threats (Black *et al.*, 2024). This security dilemma could be further exacerbated if AI-controlled AWS lower the threshold for triggering conflict by creating the illusion of lower risk to human forces, potentially leading to more frequent and more aggressive military engagements (Russel, 2023).

3.3.3 Shifts in Global Military Balance

As discussed in Section 3.2, AI-controlled AWS are already fundamentally changing military power projection at the operational level by enabling more persistent, scalable, and adaptable military operations. In contrast to conventional armed forces, they can operate continuously and without fatigue, sustain extended operations with less logistical effort, and respond dynamically to evolving threats in real time (Gunzinger and Autenreid, 2020). These capabilities will also have an impact at the strategic level, enabling states to project power beyond their traditional spheres of influence with minimal risk to personnel. The increasing integration of AI-controlled AWS in multi-domain operations could even challenge the supremacy of existing military superpowers. Countries that may not have previously been able to compete with traditional military capabilities could use AI-controlled AWS to disrupt established power hierarchies. The development of AI-controlled drone swarms and autonomous USVs raises critical concerns about the future of strategic military dominance (Chapple, 2024; Keller, 2025).

The US military's Skyborg program, which aims to integrate AI-controlled UAVs into manned combat missions, clearly shows how AI-controlled AWS can already be used as a multiplier for military operations under human control and instruction (Gunzinger and Autenreid, 2020). Conversely, opposing nations could attempt to equalize AI-dominated militaries by using asymmetric tactics such as cyberwarfare, AI-controlled misinformation, and electronic warfare to disrupt the communication and decision-making of AI-controlled AWS (Panella, 2025). In this way, the use of AI-controlled AWS also creates ethical and humanitarian challenges in terms of accountability in warfare. As AWS become more autonomous, the challenge of determining legal accountability for war crimes, unintended civilian casualties, or rogue AI-controlled attacks becomes more complex (Docherty, 2015). Without clear accountability mechanisms, the proliferation of AI-controlled AWS may lead to a strategic environment where accountability for lethal actions is difficult to assign, further complicating the laws of war (Bode and Bhila, 2024).

3.4 Impact on Military Doctrine

AI-controlled AWS require not only a legal framework for their development and deployment, but also an adaptation of military doctrine to take into account their impact on all three levels of warfare. Since AI-controlled AWS accelerate the pace of decision-making beyond human levels, they undermine traditional military structures of command and control. Military doctrine relies on the deliberate sequencing of actions within the OODA loop, with human judgment playing a significant role at each phase. This loop is still used by AI-controlled AWS, but it is greatly accelerated because it is processed at machine speed, bypassing layers of human control and decision-making. A strong criticism here is that removing humans from critical decision points can lead to flawed or even absent escalation management (Horowitz, 2019; Johnson, 2020a). This criticism has a particularly significant impact on doctrines designed for step-by-step conflict resolution and based on anticipation, signalling, and human intent, especially when AI-controlled AWS are so autonomous that they can interpret and attack threats independently within seconds. This doctrinal misalignment increases the risk of unintended escalation and undermines strategic stability (Futter, 2022; Wood, 2024).

In addition to the impact on the time factor, AI-controlled AWS also influence the spatial and functional structures anchored in traditional doctrine. To better control the growing complexity, modern military operations are structured at the tactical, operational and strategic levels, with each level having its own objectives, instruments and chains of command. AI-controlled AWS can disrupt these structures with multi-domain capabilities by achieving effects that go beyond the intended scope. For example, a tactically deployed AI-controlled AWS can impair a state's nuclear early warning system by destroying a radar system, thereby causing strategic consequences. While this blurred doctrine of levels did not come about only with the use of AI-controlled AWS, it does influence their existing framework for escalation control, proportionality and legal accountability (Bode and Huelss, 2022; Black *et al.*, 2024). Without a clear separation between the command and control levels for the use of AI- controlled AWS, commanders could have difficulty anticipating or mitigating the effects of second-order autonomous operations.

The increasing deployment of AI-controlled AWS will also make combined operations of NATO and other alliances more difficult, as these require interoperability, not only in terms of equipment but also of doctrine. Many national approaches still differ considerably, which is emphasised by the lack of consensus on the regulation of AI-controlled AWS. While some states, such as France, insist on the exclusive use of human-in-the-loop AWS, other states also allow human-on-the-loop AWS and even accept human-out-of-the-loop AWS under certain conditions, such as the USA (Boulanin, 2020; Solovyeva and Hynek, 2023; US DoD, 2023). These differences can lead to incompatible rules of engagement, operational rhythms and accountability frameworks during combined operations, which in turn endangers the unity of command. In the future, it will be imperative to find common rules of engagement for the use of AI-controlled AWS, because otherwise tactical decisions made by the AI- controlled AWS of one nation could have diplomatic consequences for an entire alliance if the supervisory mechanisms are misaligned (Atkinson, 2024; Erskine and Miller, 2024).

This source of error is amplified by the increasing reliance on AI-supported systems for operational planning and decision-making on the battlefield. They already assist with logistics, target acquisition, and threat prioritization. While this increases operational efficiency, it also increases the risk of automation bias, where under pressure, operators or commanders accept an AI-made decision without proper scrutiny (Bode and Bhila, 2024; Hoffman, 2024). Through this process, the locus of human agency shifts to algorithmic logic, slowly displacing the commander as critical thinker and ethical agent. AI-controlled AWS do not possess the contextual awareness and value-based thinking to interpret gestures of surrender, understand cultural nuances or meaningfully weigh up collateral damage. The ethical and moral cornerstone on which many modern military doctrines are based is thus called into question and will be increasingly difficult to enforce in the future (Asaro, 2020; Wood, 2024).

This shift also threatens the professional identity of soldiers, because military values anchored in doctrine, such as honour, responsibility, and discipline, are closely linked to human judgment and accountability. As AI- controlled AWS play an increasingly important role on the battlefield in the future, there is a risk that the soldier will become a passive technician or observer. This transformation could dilute military ethics and foster a culture of disconnection (Solovyeva and Hynek, 2023; Ferl, 2024). This is associated with important tools for adapting and refining military doctrines, such as lessons learned or debriefings, since these depend on statements, reports and judgments by soldiers. However, the data logs created by AI-controlled AWS require technical interpretation, which shifts the feedback loop for doctrines into the realm of software engineers and data analysts (Simmons-Edler *et al.*, 2024). To counteract this change, not only does the doctrine need to be adapted, but institutional competencies must also be changed by incorporating new training frameworks that prepare commanders for hybrid decision environments. That means that future commanders will be trained to collaborate with AI systems, not as passive observers, but as active, informed decision-makers capable of questioning or overriding algorithmic decisions. This requires doctrinal schools to expand beyond traditional operational command and to include supervision of machine reasoning, trust calibration, and ethical intervention strategies (Johnson, 2024). Additionally, maintaining meaningful human control in autonomous warfare depends on equipping soldiers with not only technical fluency, but also the ethical and legal literacy to assess AI-made decisions under pressure (Ferl, 2024). Without these new institutional competencies at all levels of warfare, the increased use of AI-controlled AWS risks demoting both commanders and soldiers to system observers in the future, instead of making professional military decisions.

3.5 Conclusion

AI-controlled AWS will shape future battlefields and therefore require not only technological superiority but also strategic responsibility, because attractive aspects of this technology, such as speed and adaptability, also make it very dangerous without appropriate or clear rules of engagement. If this trend towards machine-controlled warfare continues unrestricted, there is a risk that escalations will no longer be a matter of political decisions, but of algorithmic probabilities. Even if human operators are increasingly removed from decision-making processes, the strategic and ethical responsibility will remain with them. The challenge will be whether military institutions and political decision-makers can live up to this responsibility and keep pace with the transformation in order to shape it before it shapes itself. Without clear ethical boundaries and robust command structures, AI-controlled AWS could undermine the very principles that underpin the laws of warfare and international security norms. This technology would then pose a risk not only because of its tactical effectiveness on the battlefield, but also because of how its deployment is understood, or misunderstood, by adversaries. Misperceptions, especially in high-stakes scenarios, could have consequences far beyond the battlefield.

The effects of AI-controlled AWS on traditional levels of warfare outlined here suggest that not only will the efficiency of warfare improve, but the nature of warfare itself will change. Their deployment will accelerate conflicts, reshape command and control structures, and blur the lines between attack and defence. In addition, traditional theories of deterrence, escalation and strategic stability will need to be re-evaluated and adapted.

This will usher in a fundamental change not only in how wars are fought, but also in how they are started and prevented.

This Section has examined and analysed the implications of the use of AI-controlled AWS for future warfare. It has become clear that technological progress is only one aspect of this technology and should not be viewed in isolation. The next Section therefore addresses the question of how states can maintain political and ethical control over systems that are increasingly capable of functioning without them. The international community must act now to ensure that AI-controlled AWS of the future become a stabilising rather than a destabilising force.

4 Regulation and Governance of AI-Controlled AWS

The integration of AI into AWS renders the current legal and ethical framework obsolete. As explained in the previous Section, AI-controlled AWS will operate more autonomously in the future and thus partly outside traditional chains of command, as their deployment can extend beyond the individual levels of warfare. Accordingly, this Section will answer the question of how the deployment of AI-controlled AWS could and should be regulated. The aim is to critically examine the regulatory and governance framework that could be applied to AI-controlled AWS and to assess whether current international efforts are sufficient in light of this rapidly developing military technology.

Unlike earlier technologies such as nuclear weapons, which were primarily developed and deployed by state actors within highly centralised systems, AI-controlled AWS pose a more complex regulatory challenge. Not only are they more accessible and scalable, but their decision-making processes are also more opaque. This opacity makes accountability more difficult and increases the risk of unintended consequences. Traditional arms control mechanisms are not readily applicable, raising doubts about whether existing legal frameworks, such as the International humanitarian law (IHL), can adequately address the unique characteristics of autonomous decision-making in warfare (Boulanin, 2020; ICRC, 2021).

To answer the question of how AI-controlled AWS could and should be regulated, this Section is divided into three areas. First, existing legal norms and proposed frameworks aimed at restricting or guiding the use of AI-controlled AWS are assessed. Second, it examines the power relations and geopolitical tensions that make international agreement on regulating the deployment of AI-controlled AWS difficult. The final sub-Section of this Section deals with the development of a tiered governance model tailored to the autonomy levels of AI-controlled AWS and traditional levels of warfare, including practical proposals for implementation at the national and international levels. The central argument is that effective regulation requires a combination of legal clarity, technological control and political will. These factors are currently limited or non-existent, meaning that the increasing deployment of AI-controlled AWS risks undermining both legal accountability and strategic stability.

4.1 Regulation, Law and Power

4.1.1 Legal Frameworks

IHL and existing international agreements, in particular the Geneva and Hague Conventions, are fundamental cornerstones for the legal regulation of warfare. However, the ever-faster development and use of AI-controlled AWS is revealing more and more gaps in these frameworks. While traditional frameworks emphasize core principles such as distinction, proportionality, and military necessity, AI-controlled AWS inherently challenge these norms due to their autonomous decision-making, which can bypass real-time human oversight and control (Asaro, 2020).

With the Convention on Certain Conventional Weapons (CCW) and the United Nations Group of Governmental Experts (UN GGE) on LAWS, important forums exist for the international debate on the regulation of AI-controlled AWS. However, since the first discussions in 2014 and later through the UN GGE in 2016, it has not been able to develop legally binding regulations, mainly due to diverging national interests and strategic considerations (Boulanin, 2020). These diverging interests are also increasingly limiting the effectiveness of the CCW framework, as it is largely based on consensus among states. Due to current geopolitical tensions, particularly between major military powers such as the United States, Russia and China, this consensus is becoming increasingly difficult to achieve, and new agreements are less and less likely (Erskine and Miller, 2024). Such new agreements on the use of AI-controlled AWS are imperative in current conflicts, such as in Gaza and Ukraine. In Gaza, for example, AI-supported target recognition and identification is already in use. Even if this data still has to be confirmed by a human, this AI-supported targeting system in a densely populated area raises critical concerns regarding compliance with IHL. A clear distinction between combatants and non-combatants cannot be ensured due to an as-yet-missing contextual understanding, which leads to an increased risk of civilian casualties and subsequent international criticism (Elbaum and Panter, 2024). In addition, the development and use of AI-controlled AWS is progressing so rapidly due to the war in Ukraine that a legal interpretation, particularly with regard to the allocation of accountability in the event of errors or war crimes, will quickly become obsolete (Chapple, 2024).

Currently, the existing legal framework lacks clear definitions of who can be held responsible for unlawful acts of AI-controlled AWS and what share of the guilt this actor bears. Until it is clear whether the responsibility should lie with the manufacturers, programmers, military commanders or the state of deployment, this accountability gap undermines the deterrent effect of the IHL as a whole (Docherty, 2015). In addition, meaningful human control of AI-controlled AWS is being discussed in the CCW and UN GGE, although there is no legal clarity on what constitutes meaningful control. This creates loopholes that allow states to deploy AI-controlled AWS while still claiming compliance with all laws and regulations (Ferl, 2024). In this way, states can gradually normalize higher levels of autonomy without fear of legal consequences. The gaps in IHL not only enable an arms race, but also inadvertently accelerate it, as states and private corporations seek an economic or military advantage by developing AI-controlled AWS before the gap is closed by robust regulation. This development is confirmed by historical patterns observed in other disruptive military technologies, such as nuclear weapons. The international legal framework for regulating these technologies often lagged behind these technological innovations (Futter, 2022). An immediate reassessment of the existing legal framework is therefore necessary to take into account the complexity of the ever faster developing AI-controlled AWS technology. If there is no international consensus on specific regulations for the development and use of AI-controlled AWS, the consequence would be an increase in international conflicts and global instability.

4.1.2 Power Dynamics

The development, deployment and regulation of AI-controlled AWS is largely controlled by various actors, who, with their diverse and partly competing interests, create a complex power dynamic. The strategic doctrines, geopolitical ambitions and technological capabilities are clearly reflected in the approaches and behaviour of the major military powers, such as the USA, Russia and China. The USA, for example, prefers a gradual regulation of AI-controlled AWS instead of strict bans. In this way, it promotes its own operational flexibility and technological innovation and seeks to ensure its military superiority in the future (Black *et al.*, 2024). The strategic competition thus ignited is causing Russia and China to shift their focus to less transparent and more flexible political positions in order to counter the military superiority of the USA. China's military modernization, in which AI systems have been integrated into strategic doctrine, serves as an example. This deliberate step indicates that China intends to improve its asymmetric capabilities and deterrence in the future (Boulanin, 2020; Kania, 2020). However, other smaller states are also playing an important role in the development, deployment and regulation of AI-controlled AWS. For example, due to Russia's numerical and technological superiority, Ukraine is increasingly reliant on the use of AI-supported and even controlled UAVs and USVs (Chapple, 2024; Keller, 2025). With the experience gained in real combat conditions and the associated ethical implications, Ukraine is positioning itself as an important actor in international discussions on the regulation of AI-controlled AWS (Bergengruen, 2024). Another actor is Israel, which on the one hand triggers international concerns due to opaque deployment criteria for AI-controlled AWS such as the Harpy drone, but on the other hand also has years of experience in the use of AI-controlled AWS for defence (IAI, 2021; D’Urso, 2022).

In addition to state actors, non-governmental organizations (NGO), such as the Campaign to Stop Killer Robots (CSKR), also play a role in shaping the international discourse on the development, deployment and regulation of AI-controlled AWS. CSKR is vehemently opposed to the deployment of AI-controlled AWS without human intervention and raises ethical concerns regarding accountability and compliance with IHL (Solovyeva and Hynek, 2023). Their influence is considerable and can increase international pressure, which is reflected in forums such as the CCW or the UN GGE. However, this influence is often limited there by geopolitical realities and different national security priorities, as many state actors place supposed strategic necessities above ethical arguments (Bode and Huelss, 2022).

Driven by commercial interests, however, economic actors such as Baykar Technologies or Palantir Technologies are also increasingly driving the development of AI systems and AI-controlled AWS, and their innovations are having an ever greater impact on military effectiveness and the dynamics on the battlefield (Bergengruen, 2024; Reisner, 2024). These commercially motivated development agendas further hamper the already complicated regulatory efforts, as technological development often outpaces international legal and ethical guidelines (Futter, 2022).

The different and partly competing agendas outlined in this Section underscore the inherent challenges in achieving effective international regulation of the development and deployment of AI-controlled AWS. While NGOs and some states, such as Austria, Brazil and Mexico, advocate strict human oversight of the use of AI-controlled AWS (Wareham, 2020), major military powers and industry stakeholders are significantly hindering consensus-building because they fear the loss of military or economic advantage (Bode and Bhila, 2024). As discussed in the previous Sections, the lack of clear regulatory standards increases strategic uncertainties and makes international agreements increasingly complex and politically explosive.

4.1.3 Proposals for Regulation

As described in Section 4.1.1., there is no agreement in the international discussion on the development, deployment and effective regulation of AI-controlled AWS, as the tensions between old strict international restrictions, such as the IHL, and flexible, non-binding political declarations are seemingly irreconcilable. From an ethical point of view, binding treaties are desired and even demanded, but not all states will comply with such treaties if credible enforcement is lacking. Without such concrete enforcement mechanisms, only simple political declarations should be made, as these quickly find broad support (Bode and Huelss, 2022) and, if successfully applied, can serve as a basis for a binding regulation in the future.

CSKR emphasizes that there can be no meaningful human control in fully autonomous systems and therefore calls, from an ethical point of view, for a legally binding treaty that would completely prohibit AI-controlled AWS (Solovyeva and Hynek, 2023). However, such a complete ban faces practical limitations. Military powers such as the USA, Russia and China would object to a multilateral or even independent inspection of compliance with binding treaties due to security and sovereignty concerns (Black *et al.*, 2024). Should the current geopolitical tensions continue in the future, unanimous agreement on a complete ban on AI-controlled AWS remains unlikely, as states do not want to be disadvantaged in the ongoing technological arms race (Boulanin, 2020). These political realities seem to be deadlocked and yet the work on a possible solution should be continued. A pragmatic solution could be an incremental regulatory framework that uses normative principles. Meaningful human control, XAI and algorithmic accountability could serve as crucial governance benchmarks and provide orientation without hindering technological progress (Ferl, 2024; Wood, 2024). These standards would, of course, still need to be defined. For example, meaningful human control means that humans must maintain sufficient oversight to ensure accountability for AI-controlled AWS, but the exact definition remains controversial (Wareham, 2020). In addition to this gap in the definition, there is no clear consensus in forums such as the UN GGE on algorithmic accountability or XAI as prevailing regulatory principles, even though their importance is emphasized by humanitarian institutions such as the ICRC (ICRC, 2021; Digital Watch Observatory, 2025). This would require all decisions made by AI systems to be so comprehensible and transparent that human operators can understand them and, if necessary, question or override them. Nathan Wood criticizes the fact that in military decision-making, it is usually too stressful or time-critical to review decisions made by AI systems (Wood, 2024). However, the current conflict in Gaza, in which Israel is using opaque AI algorithms to identify targets, highlights the need for this transparency standard in military AI deployment (Elbaum and Panter, 2024), at least from the operational level.

Another proposal is to make greater use of political declarations instead of binding legal treaties. Within the framework of the United Nations, such political declarations and statements are already frequently used as a means of quickly gaining international support and creating a common consensus on a particular course of action (UNODA, 2023). However, one of the main criticisms is that these declarations are non-binding, as they are more of a symbolic gesture without any legal effect. States can therefore subscribe to them superficially without really pursuing an active political change and without fearing legal consequences if they change their minds (Erskine and Miller, 2024).

In view of the challenge of defining an internationally binding regulatory framework, a hybrid approach combining legally binding components with flexible standards could be a politically acceptable solution. This hybrid approach, which can also be referred to as incrementalism (Etzioni and Etzioni, 2017), allows for a stepwise tightening of restrictions based on demonstrated risks, technological developments and evolving ethical standards (Bode and Bhila, 2024). This could, for example, be a compliance system that requires very strict regulation at the strategic level for AI-controlled AWS to ensure control and accountability, but allows for more flexible standards for less risky, tactical AI-controlled AWS. The deployment of human-on-the-loop drones in Ukraine shows that rigid regulatory structures could impair the tactical flexibility that is essential for national defence and highlights the practical necessity of these differentiated levels of oversight (Johnson, 2022; Chapple, 2024). As incrementalism is criticised for may legalize AI-controlled AWS, this approach continues to face the challenge of reconciling ethical imperatives with geopolitical pragmatism (Asaro, 2012; Wareham, 2020). However, it also offers the possibility of regulating the development and deployment of AI-controlled AWS in a way that mitigates risks while maintaining the operational agility necessary in complex combat environments.

4.2 Tiered Governance Model for AI-Controlled AWS

4.2.1 Model Overview

The governance model proposed here for the use of AI-controlled AWS is based on the three classic levels of warfare and links these to a specific level of autonomy and the associated requirements for human control.

At the tactical level, the use of AI-controlled AWS, such as swarm drones or loitering munitions, is possible up to the highest level of autonomy, as immediate and decentralized responses are crucial at this level. Without the use of human-out-of-the-loop AWS, decisions could be delayed, which in the future could endanger speed and survivability and thus compromise tactical success (Freedberg, 2023; Chapple, 2024). This high level of autonomy can be justified by the fact that tactical operations are often limited in time and location and therefore have only limited strategic impact. This means that tactical effectiveness can be significantly increased without the risk of escalation.

At the operational level, AI-controlled AWS are used in much larger areas of operation and for longer periods of time, which can lead to an increased risk of unintended escalation or retaliation by the enemy. AI-controlled AWS at this level, such as air defence systems, must therefore be partially autonomous and thus at least deployed as human-on-the-loop systems (Johnson, 2022). An AI-based decision-making military system under human control remains flexible while ensuring that escalation thresholds are not accidentally exceeded, and that accountability can be clearly traced.

At the strategic level, the decisions and actions of AI-controlled AWS can have existential consequences, as misjudgements could destabilize entire regions or provoke nuclear retaliation. AI-controlled AWS at this level, such as nuclear missile systems or other long-range missile systems, may therefore only be deployed with minimal autonomy and only as human-in-the-loop systems. Human intuition and experience remain essential here to prevent escalation and create space for diplomatic intervention (Boulanin, 2020; Johnson, 2024).

Warfare Level	Autonomy Level	Human Control	Example of AI-controlled AWS
Tactical	High autonomy	Human-out-of-the-loop allowed	Swarm drones, Loitering munitions
Operational	Partial autonomy	Human-on-the-loop required	Air defence systems
Strategic	Minimal autonomy	Human-in-the-loop mandatory	Nuclear deterrence systems, Intercontinental missile systems

Table 1: Tiered Governance Model for Human Oversight of AI-controlled AWS

This model attempts to regulate the use of AI-controlled AWS in a simple way and deliberately avoids technical and ethical requirements. For technical models that take into account sensor fusion or decision-making speed, for example, it is not possible to establish common standards because the underlying technologies are developing too quickly and differently in different countries. In addition, no country will disclose even part of the technological progress it has made for fear of losing economic and military advantages (Boulanin, 2020; Podar and Colijn, 2025). An ethical model would also be inadequate. Ethical debates focus mainly on the permissibility of transferring lethal decisions to machines (Asaro, 2020; Russel, 2023), but often overlook the military necessity of speed, survivability, and decentralized action in modern warfare. Of course, the introduction of an ethical regulatory model would be desirable, but an agreement is highly unlikely given the ongoing geopolitical tensions, as evidenced by the deadlocked discussions within the UN GGE and the CCW (Black *et al.*, 2024; Erskine and Miller, 2024).

Despite these difficulties, a governance system for AI-controlled AWS is urgently needed. Without structured oversight, the risks of accidental escalation, accountability gaps, and the erosion of international humanitarian law will only increase with the growing proliferation of this technology (ICRC, 2021; Futter, 2022). By structuring governance according to levels of war, autonomy is only permitted where the consequences are manageable, while human intervention remains mandatory where the consequences could be existential.

4.2.2 Fluidity of Warfare Levels

Even though the tiered governance model presented in Section 4.2.1 offers a seemingly simple approach to regulating AI-controlled AWS at the tactical, operational, and strategic levels of warfare, practical implementation will become increasingly difficult in the future. In current conflicts, technological convergence, multi-domain operations, and decentralized decision-making are already leading to increasing fluidity between the levels of warfare.

AI-controlled AWS are already capable of achieving effects that can have an impact on military and other aspects. A single tactical deployment, for example on a radar system to disable local air defence, can also have strategic consequences by putting a country's early warning system out of action (Bode and Huelss, 2022; Futter, 2022). The networking of modern military infrastructures and the use of dual-use technologies make the expansion of tactical operations into political, economic, and strategic areas increasingly likely, effectively blurring the lines between the different levels of warfare (Gray, 1999). In addition, the multi-domain deployment of AI-controlled AWS on land, in the air, and at sea will lead to further overlap in command and responsibility structures (Atkinson, 2024). For example, an AI-controlled swarm of drones with minimal human intervention could support both tactical missions and operational campaigns while influencing strategic stability through autonomous data exchange and decision-making (Rashid and Khan, 2022).

Due to these developments, the governance model must adapt to a reality in which a strict separation of the levels of warfare is no longer tenable. The table in this Section refines the definitions of tactical, operational, and strategic warfare by introducing quantifiable criteria such as radius of effect, command authority, time horizon, and escalation sensitivity. This framework allows for a more nuanced application of governance principles that take current military realities into account. If even a single criterion corresponds to a higher level of warfare, the degree of human control corresponding to that higher level must be applied. This assessment principle ensures that the potential consequences of a decision, such as the use of fully autonomous AI-controlled AWS at the tactical level, are identified in advance. This makes the decision traceable and allows responsibility to be clearly assigned, which supports accountability in complex combat environments.

Criterion	Tactical Level	Operational Level	Strategic Level
Human Control	Human-out-of-the-loop allowed	Human-on-the-loop required	Human-in-the-loop mandatory
Effect Radius	Local (e.g., battlefield, urban district)	Regional (e.g., front line)	National / Global
Command Authority	Platoon / Company / Battalion	Brigade / Division / Joint Task Force	National Command Authority
Time Horizon	Immediate (minutes to hours)	Short-term (days to weeks)	Long-term (months to years)
Escalation Sensitivity	Low (short-term tactical risk)	Medium (operational blowback or enemy retaliation)	High (risk of war expansion, use of WMD[1]s, geopolitical consequences

Table 2: Warfare Levels and Required Governance Conditions for AI-Controlled AWS

4.2.3 Implementation Proposals

Currently, there are no military or legal regulations governing the use of AI-controlled AWS. For this reason, the governance model presented in the previous Sections can only be implemented by not only creating a regulatory framework, but also by adapting existing frameworks to this new technology in their entirety. However, this requires practical implementation strategies, technical security measures and institutional acceptance. To facilitate effective and ethically sound implementation, the following measures are proposed:

1) The model should be institutionalised in military doctrines at the national level and integrated into the rules of engagement (ROE). Before AI-controlled AWS

[1] WMD: Weapon of mass destruction

can be deployed, an assessment must first be carried out using the criteria for levels of warfare described in Section 4.2.2 (Effect Radius, Command Authority, Time Horizon and Escalation Sensitivity) and, if necessary, adapted to the model. Only by applying this new model, AI-controlled AWS can be used in a safe and responsible manner. This ensures a precautionary approach with regard to accountability and strategic stability (Ferl, 2024).

2) Only through effective controls the misuse of AI-controlled AWS can be prevented or punished. For this reason, monitoring of implementation by a supervisory authority is recommended. Ideally, this should be a combined military-legal supervisory authority with ethical advice. In particular for the use of AI-controlled AWS in high-risk areas such as nuclear contexts, this authority should validate the level assessments prior to deployment and issue deployment-specific approvals (Erskine and Miller, 2024). At the international level, this could be coordinated through the NATO doctrine for allied states or within UN mandates under a common legal framework. If, due to existing disagreements on the deployment of this technology, no agreement can be reached on the establishment of an authority, an authority should be introduced within an alliance of the willing.

3) The technical safety precautions for the use of AI-controlled AWS must be adapted to the various levels of autonomy and thus to the various levels of warfare. At the tactical level, where AI-controlled AWS can be used without human intervention, this means integrating automated ethical restriction modules and geofencing technologies to avoid unintended civilian casualties. In addition, fail-safe mechanisms such as dynamic kill switches, real-time anomaly detection and mission termination protocols should be standardised (Etzioni and Etzioni, 2017; Rashid and Khan, 2022). At the operational level, AI-controlled AWS must include real-time reporting mechanisms and maintain channels for human override to ensure consistency with overall campaign objectives. Human-on-the-loop must not be symbolic but must be functionally meaningful and requires training systems that enable commanders and operators to interpret, question and

override autonomous decisions (Johnson, 2022). At a strategic level, AI-controlled AWS should remain under strict human control. Their deployment must be accompanied by scenario planning prior to deployment, independent monitoring and robust chains of command to prevent misuse or misjudgement (Boulanin, 2020).

4) With regard to accountability, AI-controlled AWS must not be more than a tool for achieving an objective. It should be treated like any other weapon system, as their use is subject to strict command structures and military ethics. Accordingly, training programmes must reflect this reality and ensure that soldiers are trained not only in the technical functions of AI-controlled AWS, but also in the consequences of their misuse. As with conventional weapons, this knowledge base means that accountability for misuse lies with the commander and operator (Wood, 2024).

5) The tiered governance model presented here is only a necessary interim solution and should not be considered final. It serves as a temporary regulatory framework to at least restrict the military deployment of AI-controlled AWS where necessary to prevent greater harm. The model must remain open to iterative refinement and may eventually be replaced by a more comprehensive, ethically grounded regulatory framework, ideally within the framework of the IHL (Asaro, 2020; Digital Watch Observatory, 2025).

4.3 Conclusion

The regulation and governance of AI-controlled AWS is no longer just a theoretical question, but a strategic necessity. These systems will determine future warfare with increased speed and algorithmic logic, and the current lack of legal clarity poses a real threat. This Section has shown that IHL, as the central set of rules governing warfare, is sufficient to regulate conventional warfare, but not when it comes to the complexity of AI-made decisions in armed conflicts. This means that the gap between technological capabilities on the battlefield and the need for regulation is growing.

Attempts to remedy critical vulnerabilities, such as the lack of accountability or the absence of guidelines for operational control, are repeatedly hampered in international bodies such as the UN GGE by geopolitical competition and disputes over definitions. While major powers do not want to be legally constrained in order to maintain their military and economic advantage, others are exploiting legal ambiguities to push ahead with the development of AI-controlled AWS within an ethically questionable but not prohibited framework. In this environment of conflicting interests, a global treaty comparable to the NPT seems unlikely in the near future (Johnson, 2020b; UNODA, 2023). Nevertheless, inaction is not an option.

To close this regulatory gap, this Section has developed a tiered governance model that links the autonomy levels of AI-controlled AWS with traditional levels of warfare. It neither restricts the technological development of AI-controlled AWS nor does it completely prohibit the use of fully autonomous AWS. Until an international binding agreement is finally concluded, this model could enable national authorities, alliances and international bodies to fulfil their ethical and moral responsibilities. Ultimately, the question is not whether AI-controlled AWS can be regulated, but whether the international community is prepared to act before their use leads to an escalation of global conflicts.

5 Conclusion

AI-controlled AWS are no longer science fiction. Their use from the trenches in Ukraine to blurred radar signatures over the Red Sea shows that they are very real and are reshaping the rules of warfare. This thesis has examined their technological progress, military utility, and the fragile regulatory framework surrounding them to ultimately answer three core questions:

4) How advanced and dangerous are AI-controlled AWS really?
5) How will the deployment of AI-controlled AWS change future warfare?
6) How can the deployment of AI-controlled AWS be regulated?

This Section summarises the most important findings, consequences and implications from answering these core questions so that they can be used as guidance by the German or Irish armed forces in dealing with AI-controlled AWS in the future.

5.1 Key Findings

Case studies in Section 2 showed that AI-controlled AWS are being used in ongoing conflicts, such as in Ukraine, Israel, and Libya. The deployment ranges from semi-autonomous drones to fully autonomous loitering munitions that operate without human supervision. This demonstrates that AI-controlled AWS are not only a future threat, but a current operational reality, the deployment of which reveals profound regulatory and ethical shortcomings. Ethical frameworks, such as IHL, have not yet been adapted to this new reality, leaving AI-made decisions about life and death in an accountability gap. To address this problem, methods have been proposed to classify this new technology ethically or technically. Ethical distinctions such as human-in-, on-, or out-of-the-loop or technical criteria such as sensor fusion or the integration of machine learning are necessary tools to help lay the foundation for tailored regulation and governance frameworks.

Section 3 highlighted the future impact of AI-controlled AWS on the various levels of warfare. While advantages such as speed, precision, and reduced risk to personnel arise at the tactical and operational levels, the use of AI-controlled AWS at the strategic level poses significant, if not existential, risks. The drastic acceleration of the OODA loop greatly reduces response times and opportunities for human decision-making, increasing the likelihood of misinterpretations and unintended escalations, especially in a nuclear context. Traditional deterrence strategies rely on human rationality and calculable actions. There is a risk that these strategies will be undermined in the future by unconventional and unpredictable AI-made decisions. To mitigate this risk, the use of technical safeguards, such as the integration of XAI or restriction-based ethical models, has been proposed, along with the adaptation of military doctrine and structures. In combination with a modern training system that prepares soldiers and commanders for hybrid human-machine operations, it will thus be possible in the future to critically question algorithmic decisions and, if necessary, take ethically sound action.

Section 4 took a closer look at the regulatory and legal gaps for the deployment of AI-controlled AWS. Although international forums such as the UN GGE and CCW have been discussing this topic for years, no binding agreements have been reached, mainly due to geopolitical rivalries and disputes over definitions. However, given the urgent need for modern armed forces to regulate AI-controlled AWS, this thesis proposes a tiered governance model that combines autonomy levels with traditional levels of warfare. While this model allows for the use of human-out-of-the-loop AWS at the tactical level, where risks are locally limited, it mandates strict human control at the strategic level, where decisions can have existential consequences. As the traditional levels of warfare continue to blur, the proposed model has been further refined in terms of radius of effect, command authority, time horizon, and escalation sensitivity. Together with structural and technical implementation proposals, this model strikes a balance between operational flexibility and ethical responsibility. It thus offers a scalable regulatory approach that is in line with military realities.

5.2 Limitations

The investigation in this thesis on the regulation of AI-controlled AWS has some limitations. On the one hand, the assessments of the technology and its application made here could soon become outdated due to the rapid pace of technological progress. Although the thesis is based on current data from reports on recent conflicts, such as the war in Ukraine, future AI-controlled AWS capabilities could go beyond current assumptions.

On the other hand, this thesis is limited by its conceptual focus on state actors. Although the risk of AI-controlled AWS being used by non-state actors was briefly described in Section 2, there is currently no concrete evidence of actual use, and further analysis is therefore necessary to clearly assess this risk. The same applies to the civil-military fusion of AI-controlled technologies in countries such as China, which could affect the dynamics of proliferation (Kania, 2020).

The final limitation relates to the governance model developed in this thesis, which, although theoretically sound, has not yet been tested in practice. Its implementation would require institutional reforms and doctrinal revisions. In the international context, this would require a consensus that seems unlikely given the current tense geopolitical environment. Practical implementation should therefore be tested by responsible actors at the national level for the time being.

5.3 Recommendations for German and Irish Armed Forces

It is strongly recommended that the tiered governance model be integrated into the doctrine of the German Armed Forces, starting with a clear regulatory distinction between the levels of warfare and the acceptable levels of autonomy. In addition, Germany should use its influence in international institutions such as the EU, NATO, and the UN to continue to push for common standards for the deployment of AI-controlled AWS. In particular, as a technologically advanced but ethically bound military actor, Germany should call for "human-in-the-loop" as a basic principle for strategic systems. These common standards can be tested and refined through joint international exercises with AI-controlled AWS under the supervision of an appropriately established control body. Germany should live up to its strong legalistic tradition and humanitarian obligations and keep the development and deployment of AI-controlled AWS among allies transparent.

For Ireland, a non-aligned but increasingly cyber-aware EU member, the implications of this thesis are more nuanced. Although Ireland does not have AI-controlled AWS, it is vulnerable to spillover effects from the use of AI-controlled AWS in nearby conflicts or in one of the many international operations in which it is involved. Ireland's strategic interest in this regard should therefore lie in supporting international regulatory efforts, particularly within the UN and the EU. In addition, the armed forces should participate in international exercises involving AI-controlled AWS and improve the training of their soldiers in the area of algorithmic accountability.

5.4 Future Research

Section 5.2 has already shown that the governance model developed in this thesis is merely a theoretical concept that has not been tested for feasibility in practice. Future research could therefore focus on empirically validating the multi-level governance model using simulations or case-based war games. Testing the model in different scenarios would help to refine it, prove its practical implementation, and ultimately increase acceptance for its introduction.

Another promising area is the ethical and psychological impact of the use of AI-controlled AWS on soldiers. Even though some aspects of this topic have already been discussed in this thesis, the impact of AI-controlled AWS on human decision-makers, notions of honour and responsibility, and combat morale should be further investigated. In this context, a longitudinal study on automation bias in a military context could provide insights into how AI-controlled AWS influence human behaviour over time.

A third avenue for future research concerns the increasingly influential role of private actors and dual-use technologies in the development of AI-controlled systems. Companies such as Palantir Technologies or Baykar Technologies are shaping military innovation through commercially developed AI capabilities that are later integrated into defence applications. Future studies should therefore examine how the complex interrelationships between the military and private sectors shape governance frameworks, blur traditional distinctions between civilian and military technologies, and complicate the regulation of autonomous systems.

5.5 Concluding Reflections

This thesis has shown that AI-controlled AWS are no longer theoretical threats, but tangible realities that are fundamentally changing the structure of modern warfare. Their use calls into question not only tactical and operational principles, but also traditional strategic concepts such as deterrence and escalation control. In addition, the existing legal framework is paralysed by current geopolitical conflicts, creating a dangerous regulatory gap. However, technological inevitability does not mean that there are no ethical alternatives. Military necessity must be balanced by legal accountability and moral foresight. In this thesis therefore a governance model was developed for regulating AI-controlled AWS that represents a pragmatic but principle-based approach to the responsible use of this technology in the future.

Ultimately, it depends on the decisions made today, whether AI-controlled AWS become instruments of stability or engines of chaos. It is the responsibility of states, alliances and institutions to ensure that the transfer of power to machines does not come at the expense of the values and norms that underpin international security. The results of this thesis are therefore a call to govern with foresight rather than merely reacting too late.

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